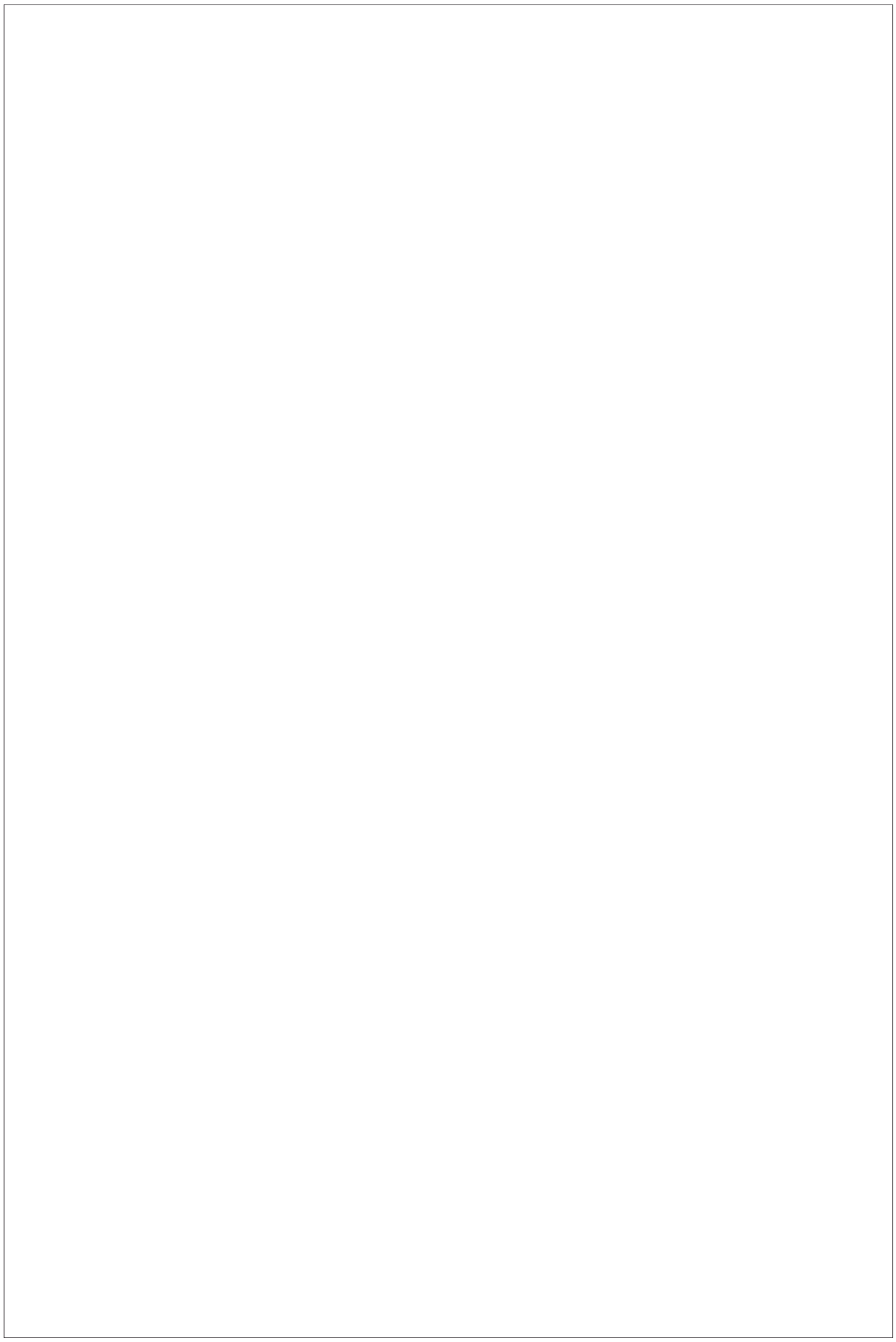
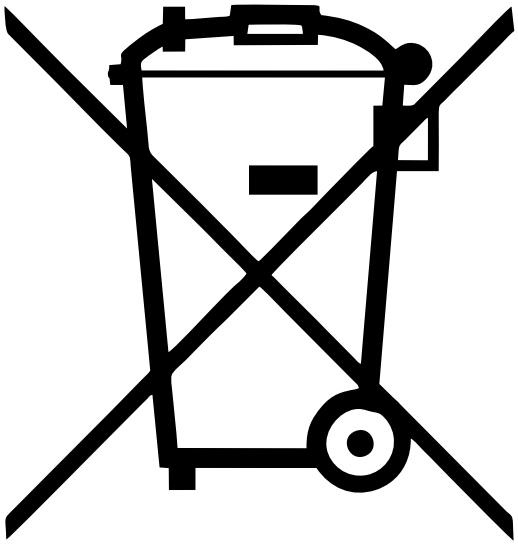
Split Type
Air-Conditioner



E- Waste Management :



This Symbol on a product and/or accompanying documents means that the used electrical and electronics products should not be mixed with general household waste. For proper treatment, recoveries, recycling, etc., please take this product to the designated collection points, where they will be accepted on a free of charge basis.

Alternatively, user may be able to return the product to Haier stores upon the purchase of a new product. Disposing off this product correctly will help save valuable resources and prevent any potential negative effect on human health and the environment, which could otherwise arise from inappropriate waste handling.

Product Features :

- Superior Durability : 100% Copper Evaporator & Condenser Coils
- Anti-Dust Filter
- Air Throw : 8 Meters

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Protection of environment



This symbol is known as the "Crossed-out Wheelie Bin Symbol". It means that the product should not be disposed of with your general house-hold waste.

It should be disposed of only through the company's collection points with special treatment so as to prevent any damage to the environment.

In line with the company's policy of continual product improvement, the aesthetic and dimensional characteristics, technical data and accessories of this appliance may be changed without notice.

Explanation of Symbols

⚠ DANGER	Indicates a hazardous situation that, if not avoided, will result in death or serious injury.
⚠ WARNING	Indicates a hazardous situation that, if not avoided, could result in death or serious injury.
⚠ CAUTION	Indicates a hazardous situation that, if not avoided, may result in minor or moderate injury.
NOTICE	Indicates important but not hazard-related information, used to indicate risk of property damage.
⚠	Indicates a hazard that would be assigned a signal word WARNING or CAUTION.

Precautions

WARNING

Operation and Maintenance

- This appliance can be used by children aged from 8 years and above and individuals with reduced physical, sensory or mental capabilities or lack of experience and knowledge if they have been given supervision or instruction concerning use of the appliance in a safe way and understand the hazards involved.
- Children should not play with the appliance.
- Cleaning and user maintenance should not be done by children without supervision.
- Do not connect the air conditioner to multi-purpose socket. It may lead to a fire hazard.
- Do disconnect power supply when cleaning the air conditioner. Otherwise, it may lead to electrical shocks.
- If the supply cord is damaged, it must be replaced by the manufacturer or their service technicians or similarly qualified persons in order to avoid a hazard.
- Do not wash the air conditioner with water to avoid electric shock.
- Do not spray water on the indoor unit. it may lead to electric shocks or malfunction.
- After removing the filter, do not touch the fins to avoid injury.
- Do not use fire or hair dryer to dry fins filter to avoid deformation or fire hazard.

Precautions

WARNING

- Maintenance must be performed by qualified professionals. Otherwise, it may cause personal injury or damage.
- Do not repair the air conditioner by yourself. It may lead to electrical shocks or damage.
- Do not extend fingers or objects into the air inlet or air outlet. It may cause personal injury or damage.
- Do not block air outlet or air inlet. It may cause malfunction.
- Do not spill water on the remote controller, otherwise the remote controller might not work properly.
- When the below phenomenon occurs, please turn off the air conditioner and disconnet the power immediately, and then contact the dealer or qualified professionals, for service.
  - Power cord is overheating or damaged.
  - There's abnormal sound during operation.
  - Circuit breaker trips off frequently.
  - Air conditioner gives off burning smell.
  - Indoor units is leaking.
- If the air conditioner operates under abnormal conditions, it may cause malfunction, electric shock or fire hazard.
- When turning on or turning off the unit by emergency operation switch, please press this switch with an insulating object.
- Do not step on the top panel of outdoor unit or put heavy objects. It may cause damage or personal injury.

Precautions

WARNING

Attachment

- Installation must be performed by qualified professionals. Otherwise, it may cause personal injury or damage.
- Kindly follow the safety regulations while installing the unit.
- According to the local safety regulations, use qualified power supply circuit and circuit breaker.
- An all-pole disconnection switch having a contact separation of at least 3mm in all poles should be connected in fixed wiring.
- Air Conditioner should be properly grounded. Incorrect grounding may cause electric shocks.
- Make sure the power supply matches with the requirement of air conditioner. Please install proper power supply cables before using the air conditioner.
- Properly connect the live wire, neutral wire and grounding wires.
- Be sure to cut off the power supply before proceeding any work related to electricity for safety.
- Do not supply power to the AC prior to finishing the complete installation.

Precautions

WARNING

- If the supply cord is damaged, it must be replaced by the manufacturer or their service agent or similarly qualified persons in order to avoid any hazard.
- The temperature of refrigerant circuit will be high, please keep the interconnection cables away from the copper tube.
- The appliance shall be installed in accordance with national wiring regulations.
- Installation must be performed in accordance with the requirement of NEC and CEC by authorized personnel only.
- The air conditioner is the first class electric appliance. It must be properly grounding with specialized grounding device by a professional. Please make sure it is always grounded effectively, otherwise it may lead to electric shocks.
- The yellow-green wire is the grounding wire, which can't be used for other purposes.
- The grounding resistance should comply with national electric safety regulations.
- The appliance must be positioned so that the plug is accessible.
- All wires of indoor unit and outdoor unit should be connected by a professional.
- If the length of any wire is insufficient, please contact the supplier for a new one. Avoid extending the wire by yourself.

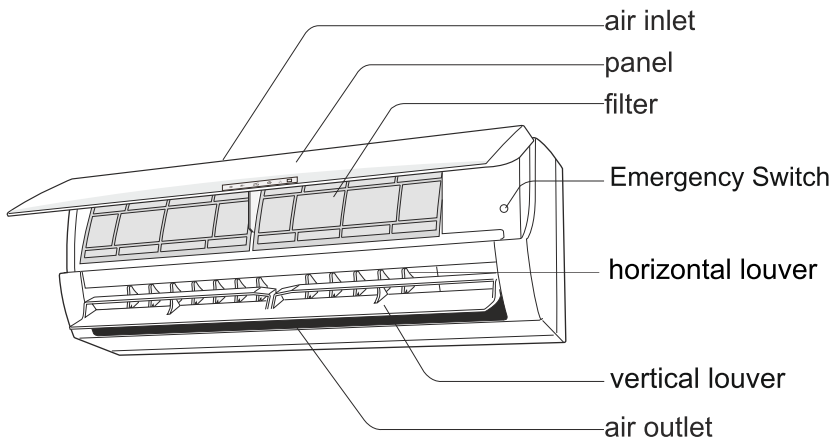
Precautions

WARNING

- For installing your Air conditioner , kindly select a location that is out of reach for children and pets. If unavoidable, kindly have a fence around the AC for safety purposes.
- The indoor unit should be installed close to the wall
- Please notice that the unit is filled with flammable gas R32. Inappropriate treatment of the unit involves the risk of severe injuries to individuals. Details about this refrigerant are found in the chapter named "refrigerant"
- Check whether the maintenance area is well-ventilated.
- Check whether there is any source of fire or potential fire source in the maintenance area.
- Check whether the appliance mark is in good condition.
  - Replace the vague or damaged warning mark.
- Please use the flammable gas detector to check before opening the container.
- The air conditioner is not allowed to be used in a room that has running fire (such as fire source, working coal gas ware, operatonal heater).
- Kindly do not drill any hole or burn the connection pipe.
- Leak test is a must after installation.
- Instructions for installation and use of this product are provided by the manufacturer.

Parts Name

Indoor Unit

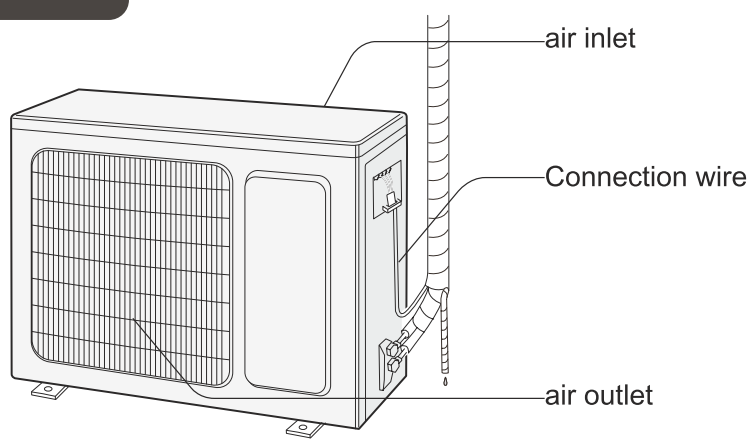


(Display content or position may be different from above graphics, please refer to actual products)



remote control

Outdoor Unit



NOTICE:

Actual product may be different from above graphics, please refer to actual products.

Parts Name

Display



“POWER” indicator : This symbol is ON when the system is in standby mode and blinks during antifungi features.

“TIMER” indicator : This symbol is ON when ON/OFF timer is activated.

“Temperature” indicator : This symbol shows the set temperature in ^{0}C in COOL/DRY/AUTO mode and room temperature in ^{0}C in FAN mode. If there is any issue in Air Conditioner Working, temperature indicator will display error codes.

“Temperature Unit” indicator : This symbol indicate the set temperature/Room temperature unit in degree Celsius.

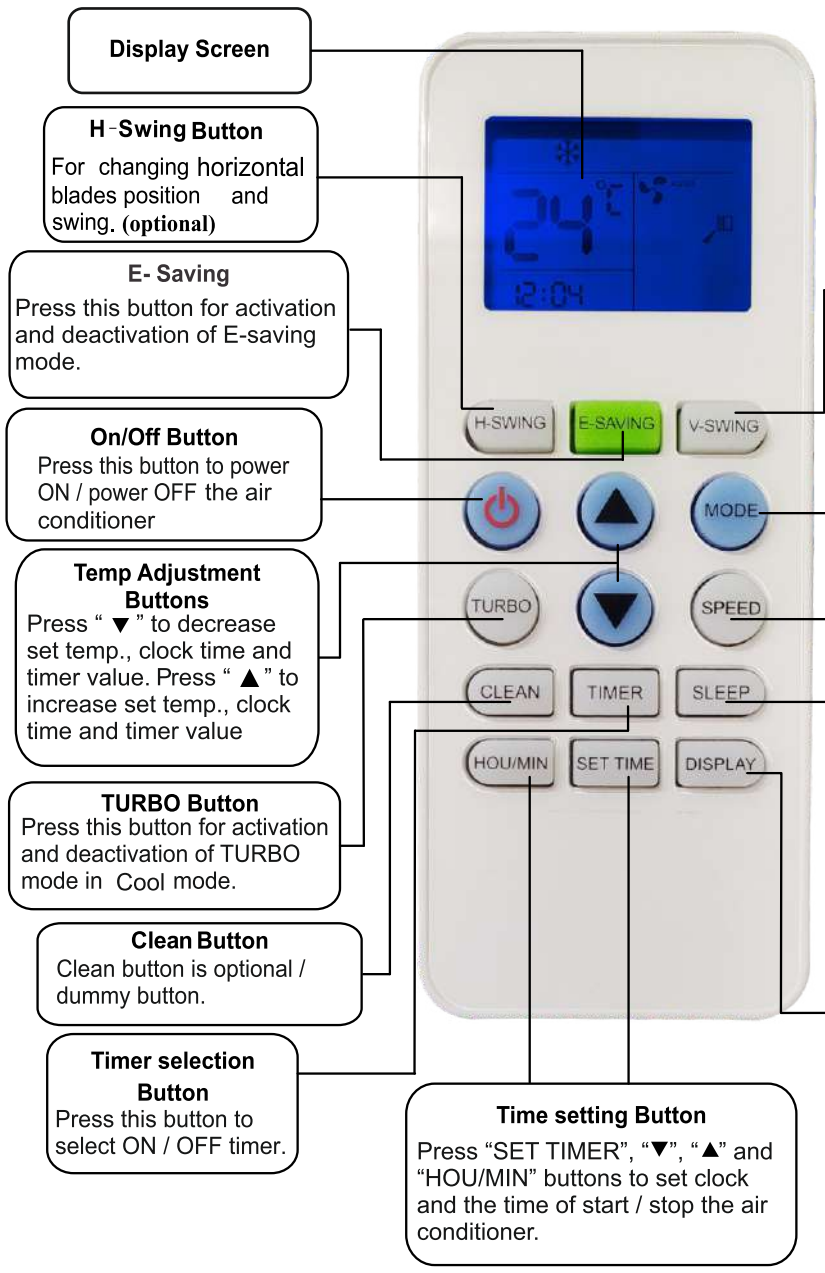
“Sleep” indicator : This symbol is ON when Sleep is activated in Cool Mode.

Remarks:

1. Function : The temperature indicator on the display screen can be controlled by “display” button on remote controller.

Display content or position may be different from above graphics, please refer to actual products.

Buttons on Remote Controller



Display Screen

H-Swing Button
For changing horizontal blades position and swing. **(optional)**

E- Saving
Press this button for activation and deactivation of E-saving mode.

On/Off Button
Press this button to power ON / power OFF the air conditioner

Temp Adjustment Buttons
Press " ▼ " to decrease set temp., clock time and timer value. Press " ▲ " to increase set temp., clock time and timer value

TURBO Button
Press this button for activation and deactivation of TURBO mode in Cool mode.

Clean Button
Clean button is optional / dummy button.

Timer selection Button
Press this button to select ON / OFF timer.

V-Swing Button
For changing vertical blades position and swing.

Mode selection Button
For selecting
COOL HEAT
FAN AUTO
DRY

Fan speed button
For selecting indoor fan speed .
Auto AUTO Low
Medium High

Sleep Button
Press this button for activation and deactivation sleep mode

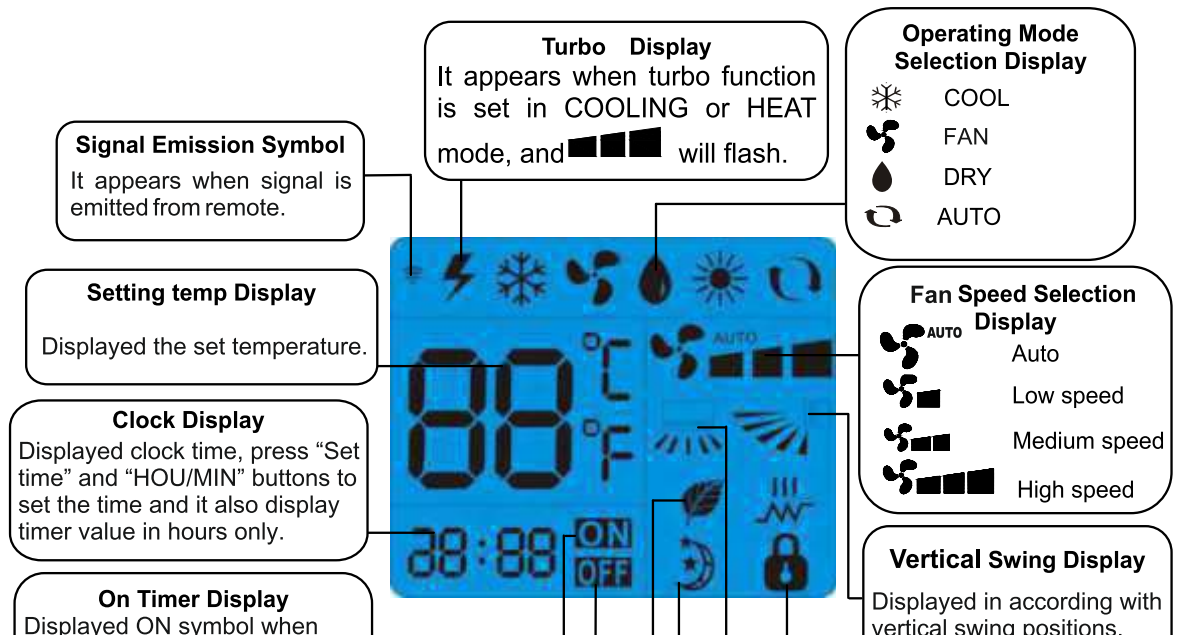
DISPLAY Button
To turn on or turn off air conditioner display when it is running as per user required condition.

Time setting Button
Press "SET TIMER", "▼", "▲" and "HOU/MIN" buttons to set clock and the time of start / stop the air conditioner.

Remarks:

1. Temperature can be set between 16^{0}C~30^{0}C.
2. The function and display of Heat is not available.
3. If user want to make the room air cool quickly, user can press"turbo" button in cool mode initially.
4. The above illustration of remote controller is only for reference, it may be slightly different from the actual product you selected.

Introduction for Icons on Display Screen



Instruction for remote controller

- The remote controller uses two AAA alkaline batteries under normal condition, the batteries last for about 6 months. Please use two new batteries of similar type (pay attention to the poles in installing).
- The effective control distance of the remote controller is about 8m.
- When using remote controller, please point the signal emitter towards indoor unit receiver; There should be no obstacle between remote controller and indoor unit.
- Pressing two buttons simultaneously may results into malfunction
- Do not use wireless equipment (such as mobile phone) near indoor unit. If interference occurs because of this, please switch off the unit, pull out power plug, then plug again and switch on after a while.
- There should be no direct sunlight to the indoor receiver, or it may not receive the signal from the remote controller.
- Don' t cast the remote controller.
- Don' t put the remote controller under the sunlight or near the oven.
- Don' t sprinkle water or other liquid on the remote controller, use soft cloth for cleaning if it occurs.
- The batteries must be removed from the appliance before it is scrapped.

Introduction for Buttons on Remote Controller

Operating Method

COOL, FAN, DRY, AUTO & TURBO operation



Switch On 1
Press On/Off button to start air conditioner.

Setting Operation Mode 2
COOL FAN DRY
AUTO

Temp Adjustment Buttons 3
Press "▼" "▲" buttons to set temp as needed

Airflow selection 4
Press "H-Swing" and "V-Swing" button to select airflow direction.

Setting Fan Speed 5
Four speeds can be selected
Auto AUTO Low
Medium High

Remarks:

1. When "COOL" mode is set, in accordance with Differential Value between indoor room temperature and set temperature, air conditioner will start or stop the cooling operation. User can change fan speed Low, Medium, High and Auto as per requirement.
2. When "FAN" mode is set, Set Temperature can't be adjusted, only Fan Speed can be adjust to Low, Medium and High. In this mode, Compressor will always remain off. Display will show indoor room temperature.
3. When "DRY" mode is set, in accordance with Differential Value between indoor room temperature and set temperature, air conditioner will start or stop the cooling operation and fan speed will change automatically to decrease room humidity. Fan speed cannot be changed in DRY mode.
   Dry Mode acts as a Dehumidifier that removes moisture from the indoor air.
   Dry Mode Refreshes air in Humid Climate. The dry function will lower the humidity and is most convenient during times in the year when it's very humid-like during the rainy season.
4. When "AUTO" mode is set, in accordance with Differential Value between indoor room temperature and set temperature, air conditioner will start or stop the cooling operation. In this, mode Set temperature become 24˚C and fan speed become Auto. Set Temperature and Fan speed can't be adjusted.
5. When "E-Saving" mode is set (in cool mode), Set temperature become 24˚C and Fan speed become Medium. Set Temperature and Fan speed can't be adjusted.
6. When "SLEEP" mode is set (in cool mode), after 5 second room temperature will disappear.
   Sleep Function will run as per following logic:
   For first hour Set temp maintains
   For next 2 hours: Set temp. + 1°C maintains
   For next 5 hours: Set temp. + 2°C maintains
   After 8 hours Sleep function gets cancelled.
7. "HEAT" mode is not applicable.
8. The Turbo feature activates by pressing the TURBO key in cool mode. IDU Fan starts rotating at High speed when pressing the TURBO key from the remote. Users can change the Set temperature (16°C to 30°C) from Remote during TURBO activation. Vertical Swing fix at F3 position for maximum airflow when pressed TURBO Key.

Introduction for Buttons on Remote Controller

Clock Setting

1. Press “Set Time” button to select the time you wish to set. The hour clock display will flash
2. Press“▼”or “▲” buttons to set hour as needed, each press of “▼”or “▲”button will decrease or increase the time display by 1 hour/1 minute; hour display will be changed continuously when pressing “▼”or “▲” continuously.
3. Press “HOU/MIN” button to change to the minute display, minute of clock display will flash. And then press “▼”or “▲” buttons to set minute as needed(the method as step 2), The hour/minute display will flash for 10 seconds after setting, the remote controller displays current time.



On Timer

1. Press “▼" or “▲“ button for set timer value
2. Press Timer button to activate and deactivate Timer function



Note:

1. When on timer function is set, air conditioner should be in OFF state. It will be turned on at the set time, symbol “**ON**” will get disappeared. If the set time is as same to the current time, air conditioner will be turned on at this time of next day.
2. Press “On/Off” to cancel “On Timer” function, air conditioner will be turned on immediately.

Introduction for Buttons on Remote Controller

Off Timer



1 Press "▼" or "▲" button for set timer value

2 Press Timer button to activate and deactivate Timer function

Note:

1. When off timer function is set, air conditioner should be in ON state. It will be turned off at the set time, symbol "**OFF**" will get disappeared. If the set time is as same to the current time, the air conditioner will be turned off at this time of next day.
2. Press "On/Off" to cancel "Off Timer" function, air conditioner will be turned off immediately.

Introduction for Buttons on Remote Controller

How to Adjust Air Flow

Horizontal direction

If press the “H-SWING” on the remote controller, horizontal louver blades can be set at the fixed position (Fig. 1) or auto swing.

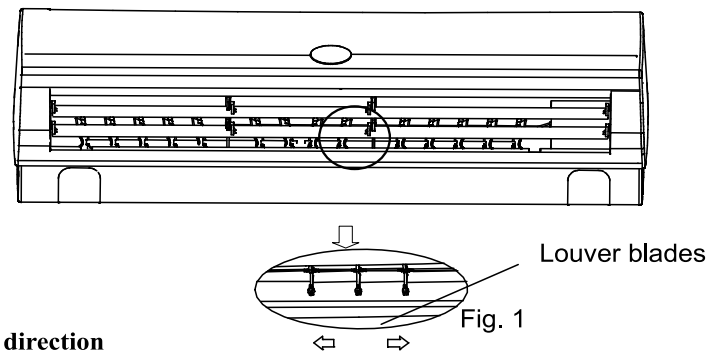


Fig. 1

Warning:
Fan inside;
Keep hands away!

Vertical direction

Press the “V-SWING” on the remote controller repeatedly, louver blades can be set at the fixed position 1-5 (Fig. 2) or auto swing activates. In different modes, louver blades swings repeatedly in the set range.

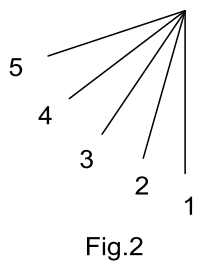


Fig.2

Sleep Operation (In Cool Mode Only).

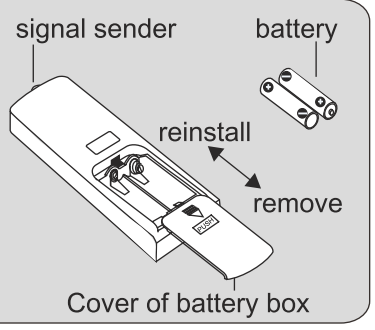
1. Press “SLEEP” button in Cool Mode to enter “SLEEP” mode and Remote will display icon.
2. Press “SLEEP” button again, icon will disappear on the remote controller, and the sleep function will be cancelled.
3. Sleep Function will run as per following logic:
   For first hour Set temp maintains
   For next 2 hours: Set temp. + 1°C maintains
   For next 5 hours: Set temp. + 2°C maintains
   After 8 hours Sleep function gets cancelled.

Note:

The set temperature will be controlled between 16^0C and 30^0C.

Replacement of Batteries in Remote Controller

1.Press the back side of remote controller marked with "PUSH", as shown in the fig, and then push out the cover of battery box along the arrow direction.
2.Replace two 7# (AAA 1.5V) dry batteries, and make sure the position of "+" polar and "-" polar are correct.
3.Reinstall the cover of battery box.



NOTICE

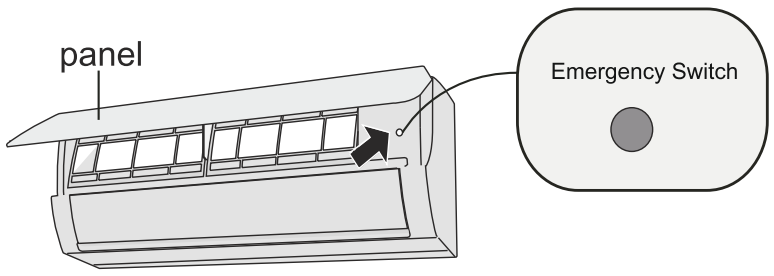
- During operation, point the remote control signal sender at the receiving window on indoor unit.
- The distance between signal sender and receiving window should be no more than 8m, and there should be no obstacles between them.
- Signal may be interfered easily in the room where there is fluorescent lamp or wireless telephone; remote controller should be close to indoor unit during operation.
- Replace new batteries of the same model when replacement is required.
- When you don't use remote controller for a long time, please take out the batteries.
- If the display on remote controller is fuzzy or there's no display, please replace batteries.

- If user has set the temperature below 24^{0}C in Cool/Dry mode prior to switching off AC and then if AC is turned on by Remote, AC will start operating at 24^{0}C temperature only.

- If user has set the temperature 24^{0}C or above 24^{0}C in Cool/Dry mode prior to switching off AC and then if AC is turned on by Remote, AC will start operating at previous set temperature set by user.

Emergency Operation

If remote controller is lost or damaged, please use Emergency Switch to turn on or turn off the air conditioner. The operation in details are as below:
As shown in the fig. Open panel, press **Emergency Switch** to turn ON/OFF the air conditioner. When the air conditioner is turned on, it will operate under auto mode.



⚠ WARNING:

Use insulated object to press the Emergency Switch.

Clean and maintenance

⚠ WARNING

- Turn off the air conditioner and disconnect the power before cleaning the air conditioner to avoid electric shock.
- Do not wash the air conditioner with water to avoid electric shock.
- Do not use volatile liquid to clean the air conditioner.

Clean surface of indoor unit

When the surface of indoor unit is dirty, it is recommended to use a soft dry cloth or wet cloth to wipe it.

NOTICE:

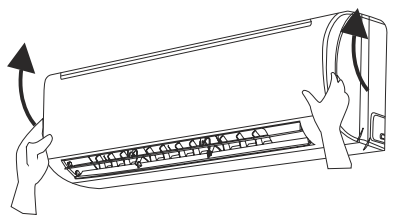
- Do not remove the panel when cleaning it.

Clean and maintenance

Clean filter

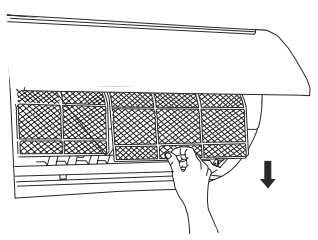
1 Open panel

Pull out the panel to a certain angle as shown in the fig.



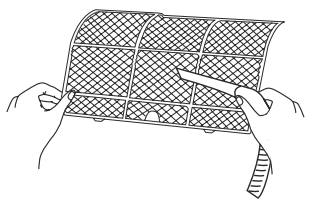
2 Remove filter

Remove the filter as indicated in the fig.



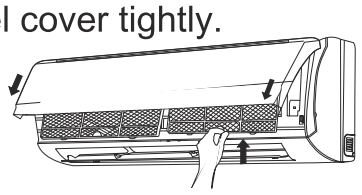
3 Clean filter

- Use dust catcher or water to clean the filter.
- When the filter is very dirty, use the water (below 45℃) to clean it, and then put it in a shady and cool place to dry.



4 Install filter

Install the filter and then close the panel cover tightly.



WARNING

- The filter should be cleaned every months. If there is much dust in the operation environment, cleaning frequency can be increased.
- After removing the filter, do not touch fins to avoid injury.
- Do not use fire or hair dryer to dry the filter to avoid deformation or fire hazard.

Clean and maintenance

NOTICE: Checking before use-season

1.Check whether air inlets and air outlets are blocked.
2.Check whether circuit break, plug and socket are in good condition.
3. Check whether filter is clean.
4.Check whether mounting bracket for outdoor unit is damaged or corroded.
If yes, please contact service provider.
5.Check whether drainage pipe is damaged.

NOTICE: Checking after use-season

1.Disconnect power supply.
2. Clean filter and indoor unit's panel.
3.Check whether mounting bracket for outdoor unit is damaged or corroded.
If yes, please contact service provider.

Notice for recovery

1.Many packing materials are recyclable materials.
Please dispose them in appropriate recycling unit.
2.If you want to dispose the air conditioner, please contact us on our Toll Free Numbers.

Malfunction analysis

Error Code

- **For Fixed Speed Model**

Code	Error Description
E1	Room Temperature Sensor short or open
E2	Coil Temperature Sensor short or open
E5/H6	Fan Motor error
E6/U8	Zero Crossing error
dF	Defrost error

- **For Inverter Model**

ODU Flash Light	IDU Error Code	Abnormality or Protection Description
25	**EE**	IDU EEPROM Fault
26	**F0**	IDU Fan motor Fault
27	**E1**	IDU PCB Zero-Crossing Fault
28	**F3**	IDU Evaporator Temperature Sensor Fault
29	**F1**	IDU Room Air Temperature Sensor Fault
1	**EF**	ODU EEPROM Fault
2	**F6**	IDU to ODU Communication Fault
4	**E4**	Compressor Starting Abnornal
5	**E3**	Compressor out-of-step Abnormal
6	**F9**	IPM Module Fault
8	**F5**	Compressor Discharge Temperature Sensor fault
9	**E5**	Compressor Suction Temperature Sensor fault
10	**F4**	Compressor Condenser Temperature Sensor fault
11	**F2**	Compressor Ambient Temperature Sensor fault
12	**E2**	Outdoor DC Fan motor fault
13	**P2**	ODU AC Current protection
14	**P0**	Compressor phase Current protection
15	**P7**	ODU over-high/over-low AC Voltage Protection
16	**P8**	ODU over-high/over-low DC Voltage Protection
17	**P9**	IPM Overheat temperature protection
18	**P1**	Compressor Discharge Temperature Overheat protection
19	**P5**	Cooling Indoor unit coil anti-freezing protection
20	**P6**	Cooling Outdoor coil Overheat protection
21	**P4**	Heating indoor coil Overheat protection
22	**PA**	Cooling outdoor ambient temperature over-low protection
23	**PE**	Heating outdoor ambient temperature over-high protection

⚠ WARNING

- When below phenomenon occurs, please turn off air conditioner and disconnect power immediately, and then contact the dealer or qualified professionals for service.
  - Power cord is overheating or damaged.
  - There is abnormal sound during operation.
  - Circuit break trips off frequently.
  - Air conditioner gives off burning smell.
  - Indoor unit is leaking.
- Do not repair or refit the air conditioner by yourself.
- If the air conditioner operates under abnormal conditions, it may cause malfunction, electric shock or fire hazard.

Malfunction analysis

General phenomenon analysis

Please check below items before asking for maintenance. If the malfunction still can't be eliminated, please contact local dealer or qualified professionals.

Phenomenon	Check items	Solution
Indoor unit cannot receive remote controller signal or remote controller has no action.	• Whether it's interfered severely (such as static electricity, stable voltage)?	• Pull out the plug. Reinsert the plug after about 3min, and then turn on the unit again.
	• Whether remote controller is within the signal receiving range?	• Signal receiving range is 8m.
	• Whether there are obstacles?	• Remove obstacles.
	• Whether remote controller is pointing at the receiving window?	• Select proper angle and point the remote controller at the receiving window on indoor unit.
	• Is sensitivity of remote controller low; fuzzy display and no display?	• Check the batteries. If the power of batteries is too low, please replace them.
	• No display when operating remote controller?	• Check whether remote controller appears to be damaged. If yes, replace it.
	• Fluorescent lamp in room?	• Take the remote controller close to indoor unit. • Turn off the fluoresent lamp and then try it again.
No air emitted from indoor unit	• Air inlet or air outlet of indoor unit is blocked?	• Eliminate obstacles.
	• Under heating mode, indoor temperature is reached to set temperature?	• After reaching to set temperature, indoor unit will stop blowing out air.
	• Heating mode is turned on just now?	• In order to prevent blowing out cold air, indoor unit will be started after delaying for several minutes, which is a normal phenomenon.

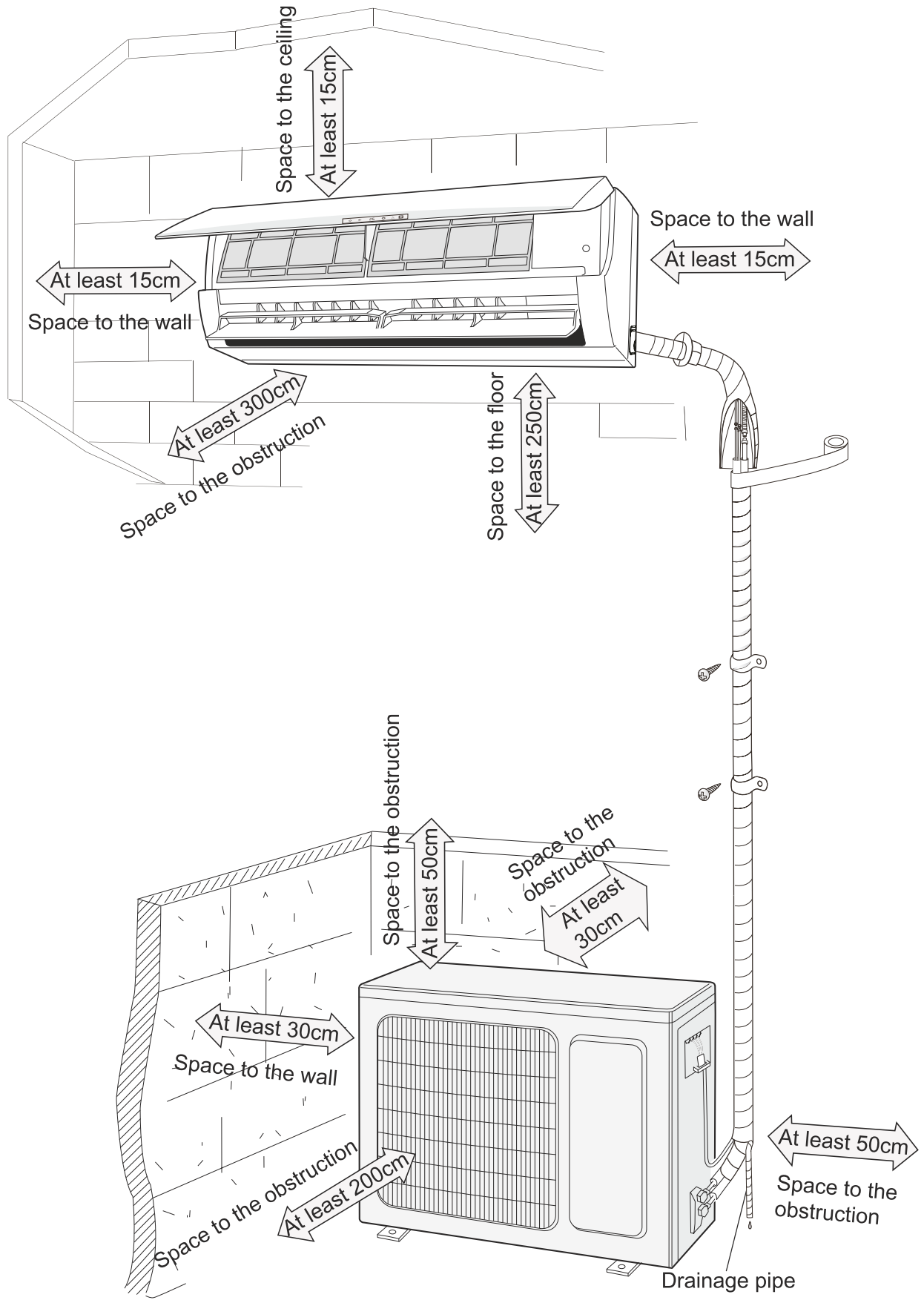
Malfunction analysis

Phenomenon	Check items	Solution
Air conditioner can't operate	• Power failure?	• Wait until power recovery.
	• Is plug loose?	• Reinsert the plug.
	• Circuit break trips off or fuse is burnt out?	• Ask professional to replace circuit break or fuse.
	• Wiring has malfunction?	• Ask professional to replace it.
	• Unit has restarted immediately after stopping operation?	• Wait for 3min, and then turn on the unit again.
	• Whether the function setting for remote controller is correct?	• Reset the function.
Mist is emitted from indoor unit's air outlet	• Indoor temperature and humidity is high?	• Because indoor air is cooled rapidly. After a while, indoor temperature and humidity will be decrease and mist will disappear.
Set temperature can't be adjusted	• Unit is operating under auto mode?	• Temperature can't be adjusted under auto mode. Please switch the operation mode if you need to adjust temperature.
	• Your required temperature exceeds the set temperature range?	• Set temperature range: 16℃ ~30℃ .
Cooling/ heating effect is not good.	• Voltage is too low?	• Wait until the voltage resumes normal.
	• Filter is dirty?	• Clean the filter.
	• Set temperature is in proper range?	• Adjust temperature to proper range.
	• Door and window are open?	• Close door and window.

Malfunction analysis

Phenomenon	Check items	Solution
Odours are emitted	• Whether there is odour source, such as furniture and cigarette, etc.	• Eliminate the odour source. • Clean the filter.
Air conditioner operates abnormally	• Whether there is interference, such as thunder, wireless devices, etc.	• Disconnect power, put back power, and then turn on the unit again.
Outdoor unit has vapor	• Heating mode is turned on?	• During defrosting under heating mode, it may generate vapor, which is a normal phenomenon.
Water flowing noise	• Air conditioner is turned on or turned off just now?	• The noise is the sound of refrigerant flowing inside the unit, which is a normal phenomenon.
Cracking noise	• Air conditioner is turned on or turned off just now?	• This is the sound of friction caused by expansion and contraction of panel or other parts due to the change of temperature.

Installation dimension diagram



Tools for installation

1 Level meter	2 Screw driver	3 Impact drill
4 Drill head	5 Pipe expander	6 Torque wrench
7 Open-end wrench	8 Pipe cutter	9 Leakage detector
10 Vacuum pump	11 Pressure meter	12 Universal meter
13 Inner hexagon spanner		14 Measuring tape

Note:

- Please contact us on our Toll Free Number.
- Don't use unqualified power cord.

Selection of installation location

Basic requirement

Installing the unit in the following places may cause malfunction. If it is unavoidable, please consult the local dealer:

1. The place with strong heat sources, vapors, flammable or explosive gas, or volatile objects spread in the air.
2. The place with high-frequency devices (such as welding machine, medical equipment).
3. The place near coast area.
4. The place with oil or fumes in the air.
5. The place with sulfurated gas.
6. Other places with special circumstances.
7. Do not use the unit in the immediate surroundings of a laundry, a bath, shower or a swimming pool.

Indoor unit

1. There should be no obstruction near air inlet and air outlet.
2. Select a location where the condensation water can be dispersed easily and won't affect other people.
3. Select a location which is convenient to connect the outdoor unit and near the power socket.
4. Select a location which is out of reach for children.
5. The location should be able to withstand the weight of indoor unit and should not increase noise and vibration.
6. The appliance must be installed 2.5m above floor.
7. Don't install the indoor unit right above the electric appliance.
8. Please try your best to keep away from fluorescent lamp.

Outdoor unit

1. Select a location where the noise and outflow air emitted by the outdoor unit will not affect neighbourhood.
2. The location should be well ventilated and dry, in which the outdoor unit won't be exposed directly to sunlight or strong wind.
3. The location should be able to withstand the weight of outdoor unit.
4. Make sure that the installation follows the requirement of installation dimension diagram.
5. Select a location which is out of reach for children and far away from animals or plants.If it is unavoidable, please add the fence for safety purpose.

Requirements for electric connection

Safety precaution

1. Must follow the electric safety regulations when installing the unit.
2. According to the local safety regulations, use qualified power supply circuit and circuit break.
3. Make sure the power supply matches with the requirement of air conditioner. Unstable power supply or incorrect wiring or malfunction. Please install proper power supply cables before using the air conditioner.
4. Properly connect the live wire, neutral wire and grounding wire of power socket.
5. Be sure to cut off the power supply before proceeding any work related to electricity and safety.
6. Do not put through the power before finishing installation.
7. If the supply cord is damaged, it must be replaced by the manufacturer, its service agent or similarly qualified persons in order to avoid a hazard.
8. The temperature of refrigerant circuit will be high, please keep the interconnection cable away from the copper tube.
9. The appliance shall be installed in accordance with national wiring regulations.
10. Installation must be performed in accordance with the requirement of NEC and CEC by authorized personnel only

Grounding requirement

1. The air conditioner is the first class electric appliance. It must be properly grounding with specialized grounding device by a professional. Please make sure it is always grounded effectively, otherwise it may cause electric shock.
2. The yellow-green wire in air conditioner is grounding wire, which can't be used for other purposes.
3. The grounding resistance should comply with national electric safety regulations.
4. The appliance must be positioned so that the plug is accessible.
5. An all-pole disconnection switch having a contact separation of at least 3mm in all poles should be connected in fixed wiring.
6. Including an circuit break with suitable capacity, please note the following table. Air switch should be included magnet buckle and heating buckle function, it can protect the circuit-short and overload. (Caution: please do not use the fuse only for protect the circuit)

Air-conditioner	Circuit break capacity
09K	10A
12K	16A
18K/24K	25A

Installation of indoor unit

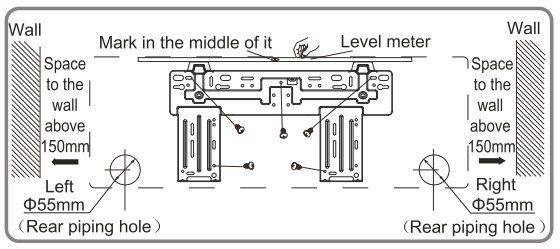
Step one: choosing installation location

Recommend the installation location to the client and then confirm it with the client.

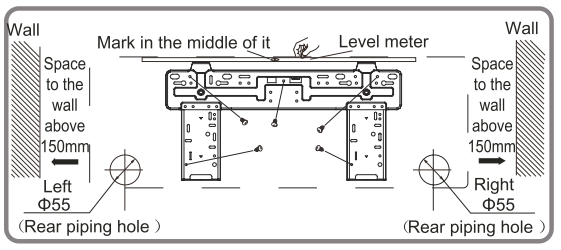
Step two: open piping hole

1. Choose the position of piping hole according to the direction of outlet pipe. The position of piping hole should be a little lower than the wall-mounted frame,

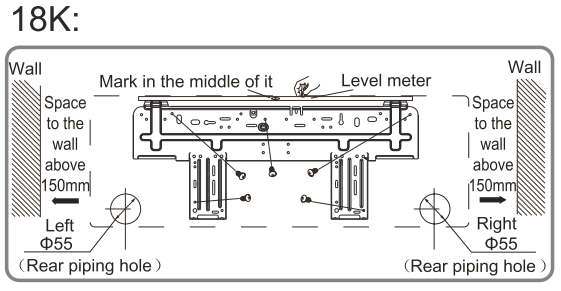
09K:



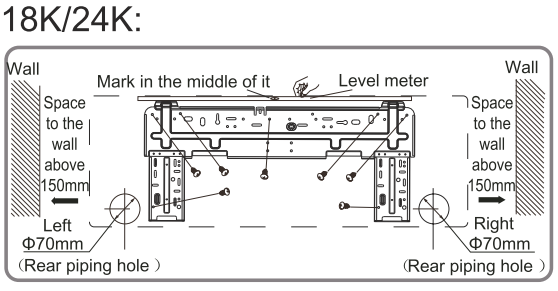
12K:



18K:



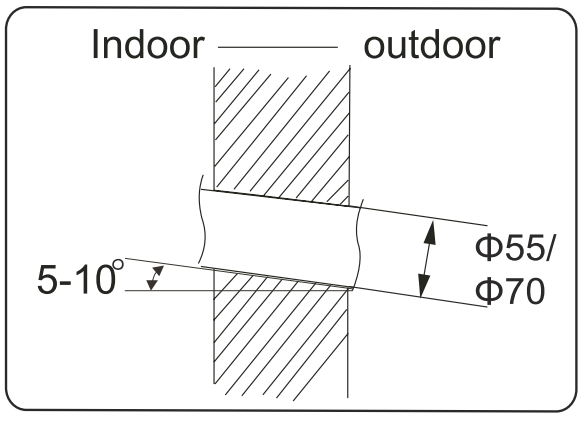
18K/24K:



2. Open a piping hole with the diameter of Φ55 or Φ70 on the selected outlet pipe position. In order to drain smoothly, slant the piping hole on the wall slightly downward to the outdoor side with the gradient of 5-10°.

Note:

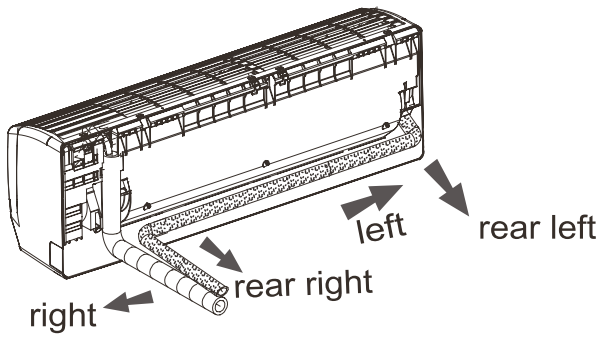
- Pay attention to dust prevention and take relevant safety measures when opening the hole.
- The plastic expansion particles are not provided and should be bought locally.



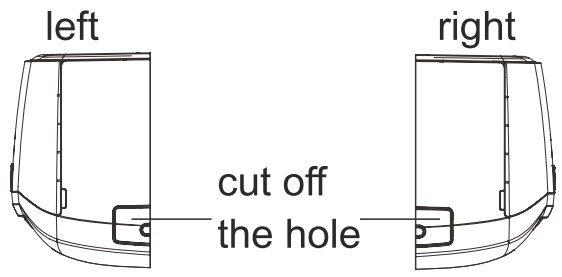
Installation of indoor unit

Step three: outlet pipe

1. The pipe can be led out in the direction of right, rear right, left or rear left.

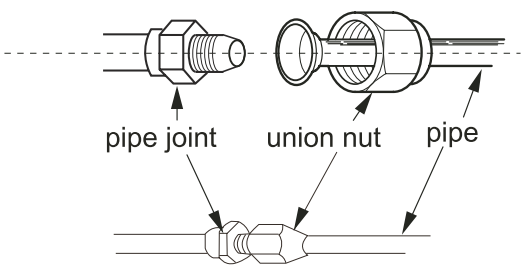


2. When select leading out the pipe from left or right, please cut off the corresponding hole on the bottom case.



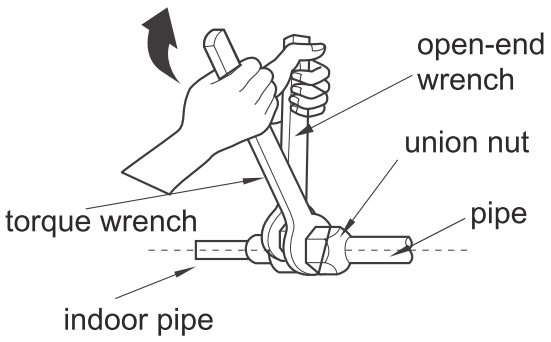
Step four: connect the pipe of indoor unit

1. Aim the pipe joint at the corresponding bellmouth.



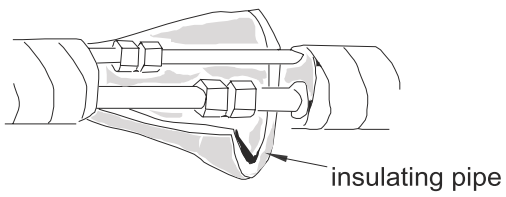
2. Pretightening the union nut with hand.

3. Adjust the torque force by referring to the following sheet. Place the open-end wrench on the pipe joint and place the torque wrench on the union nut. Tighten the union nut with torque wrench.



Hex nut diameter	Tightening torque (N·m)
Φ 6.35	15~20
Φ 9.52	30~40
Φ 12.7	45~55
Φ 15.88	60~65

4. Wrap the indoor pipe and joint of connection pipe with insulating pipe, and then wrap it with tape.

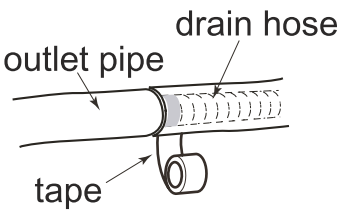


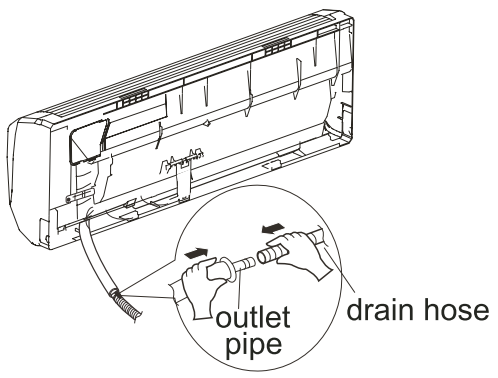
Installation of indoor unit

Step five: install drain hose

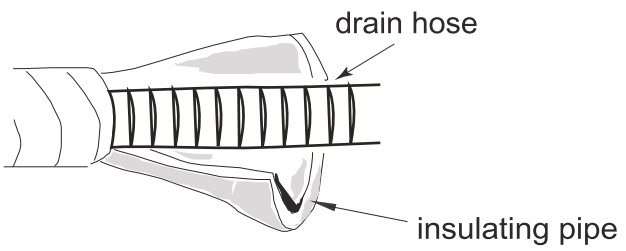
1. Connect the drain hose to the outlet pipe of indoor unit.

2. Bind the joint with tape.



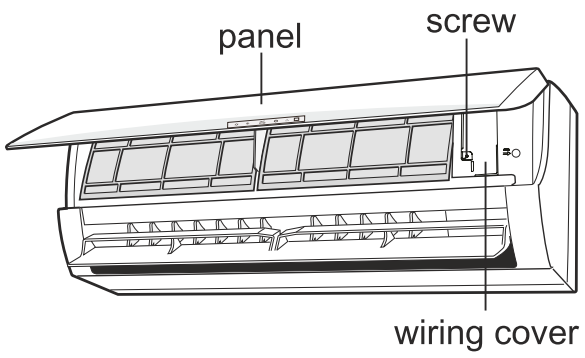


Note:
- Add insulating pipe in the indoor drain hose in order to prevent condensation.
- The plastic expansion particles are not provided.



Step six: connect wire of indoor unit

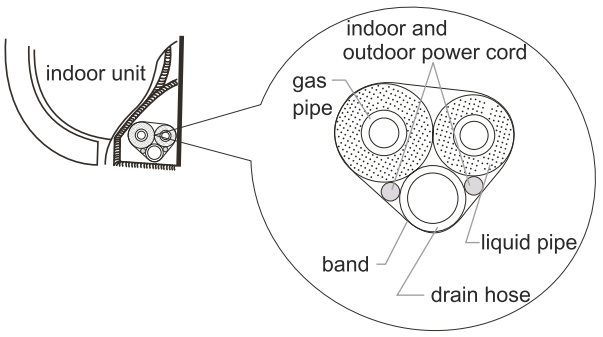
1. Open the panel, remove the screw on the wiring cover and then take down the cover.



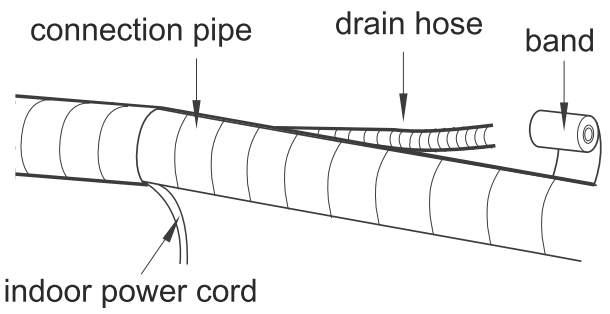
Installation of indoor unit

Step seven: bind up pipe

1. Bind up the connection pipe, power cord and drain hose with the band.



2. Reserve a certain length of drain hose and power cord for installation when binding them. When binding to a certain degree, separate the indoor power and then separate the drain hose.

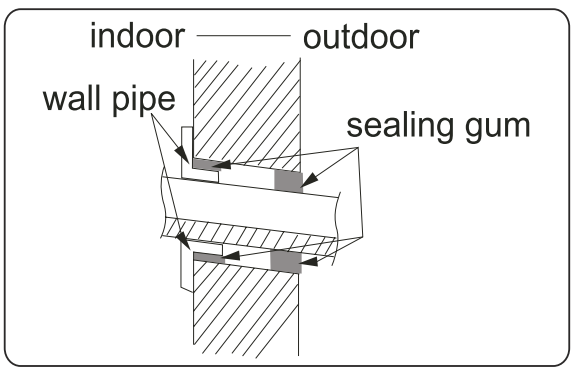


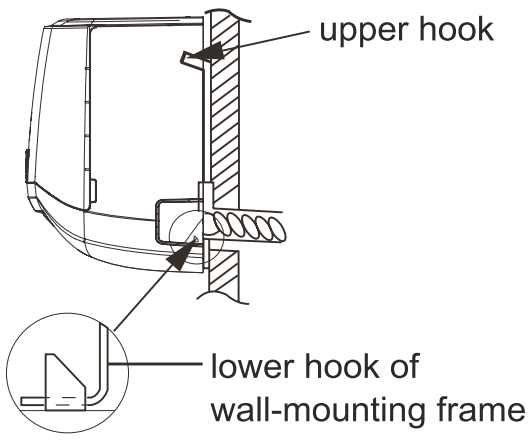
3. Bind them evenly.
4. The liquid pipe and gas pipe should be bound separately at the end.

Note:
- The power cord and control wire can't be crossed or winding.
- The drain hose should be bound at the bottom.

Step eight: hang the indoor unit

1. Put the bound pipes in the wall pipe and then make them pass through the wall hole.
2. Hang the indoor unit on the wall-mounting frame.
3. Stuff the gap between pipes and wall hole with sealing gum.
4. Fix the wall pipe.
5. Check if the indoor unit is installed firmly and closed to the wall.





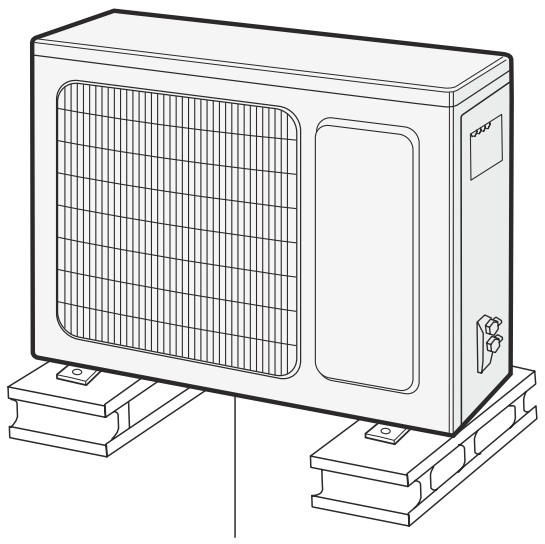
Note:
- Do not bend the drain hose too excessively in order to prevent blocking.

Installation of outdoor unit

Step one: fix the support of outdoor unit
(select it according to the actual installation situation)

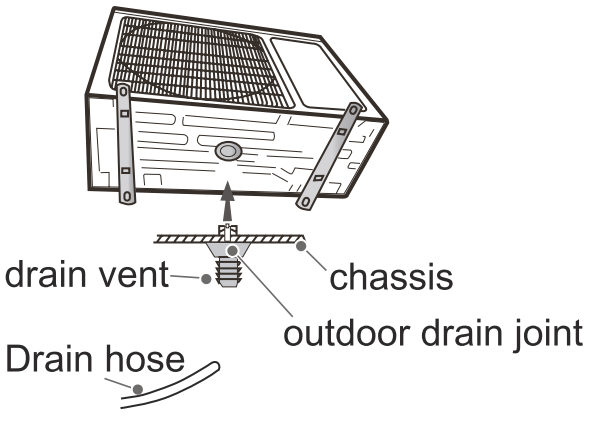
1. Select installation location according to the house structure.
2. Fix the support of outdoor unit on the selected location with expansion screws.

Note:
- Take sufficient protective measures when installing the outdoor unit.
- Make sure the support can withstand at least four times of the unit weight.
- The outdoor unit should be installed at least 3cm above the floor in order to install drain joint.
- For the unit with cooling capacity of 2300W ~5000W, 6 expansion screws are needed; for the unit with cooling capacity of 6000W ~8000W, 8 expansion screws are needed; for the unit with cooling capacity of 10000W ~16000W, 10 expansion screws are needed.



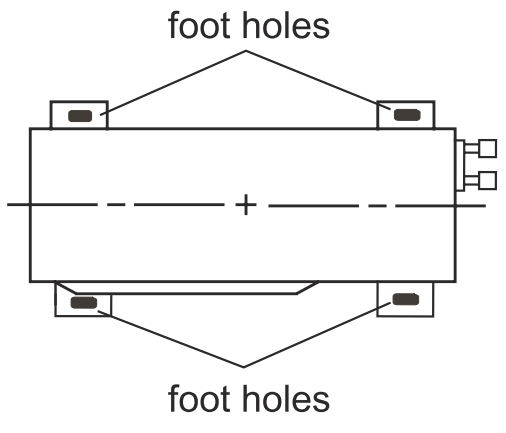
Step two: install drain joint
(Only for cooling and heating unit)

1. Connect the outdoor drain joint into the hole on the chassis, as shown in the picture below.
2. Connect the drain hose into the drain vent.



Step three: fix outdoor unit

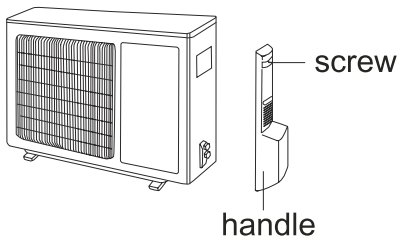
1. Place the outdoor unit on the support.
2. Fix the foot holes of outdoor unit with bolts.



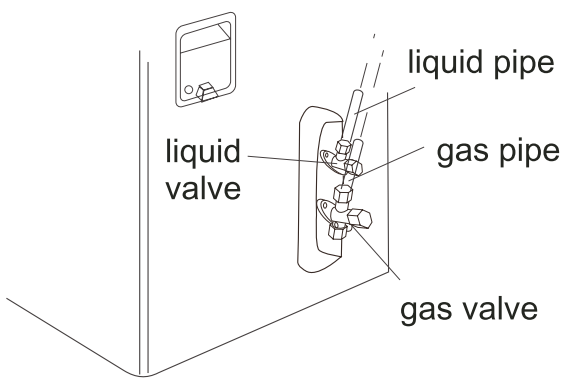
Installation of outdoor unit

Step four: connect indoor and outdoor pipes

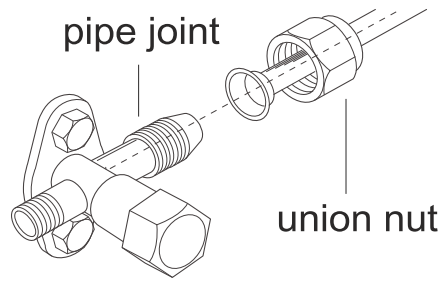
1. Remove the screw on the right handle of outdoor unit and then remove the handle.



2. Remove the screw cap of valve and aim the pipe joint at the bellmouth of pipe.



3. Pretightening the union nut with hand.

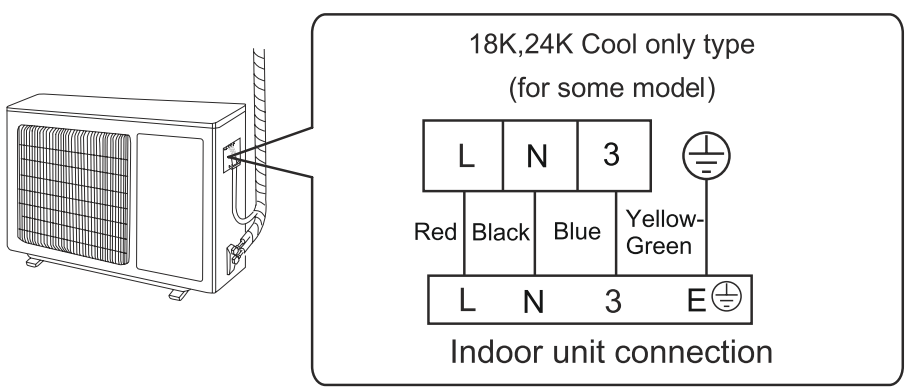


4. Tighten the union nut with torque wrench by referring to the sheet below.

Hex nut diameter	Tightening torque (N·m)
Φ 6.35	15~20
Φ 9.52	30~40
Φ 12.7	45~55
Φ 15.88	60~65

Step five: connect outdoor electric wire

1. Remove the wire clip; connect the power connection wire and signal control wire (only for cooling and heating unit) to the wiring terminal according to the color; fix them with screws.



Installation of outdoor unit

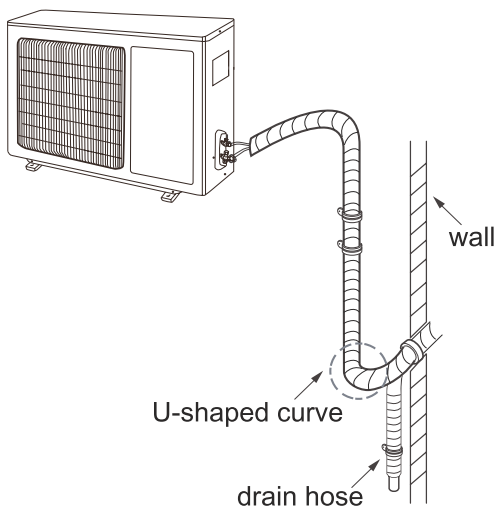
2. Fix the power connection wire and signal control wire with wire clip (only for cooling and heating unit).

Note:
- After tighten the screw, pull the power cord slightly to check if it is firm.
- Never cut the power connection wire to prolong or shorten the distance.

Step six: neaten the pipes

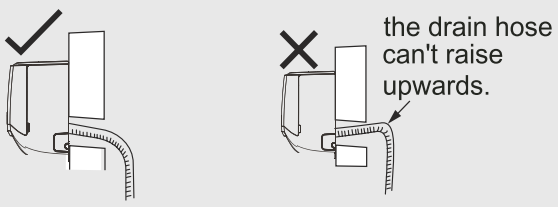
1. The pipes should be placed along the wall, bent reasonably and hidden possibly. Min. semidiameter of bending the pipe is 10cm.

2. If the outdoor unit is higher than the wall hole, you must set a U-shaped curve in the pipe before pipe goes into the room, in order to prevent rain from getting into the room.

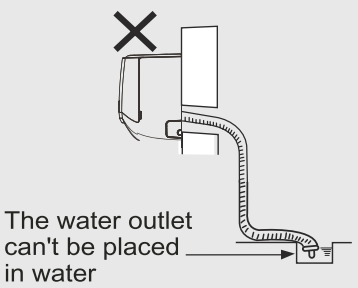


Note:

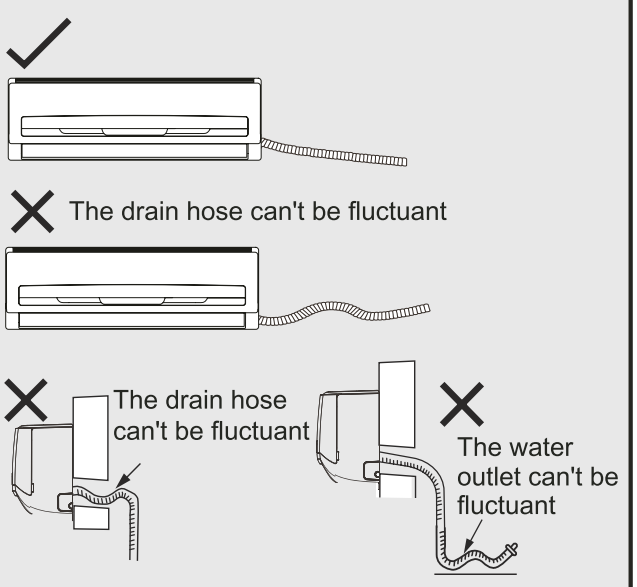
- The through-wal height of drain hose shouldn't be higher than the outlet pipe hole of indoor unit.



- The water outlet can't be placed in water in order to drain smoothly.



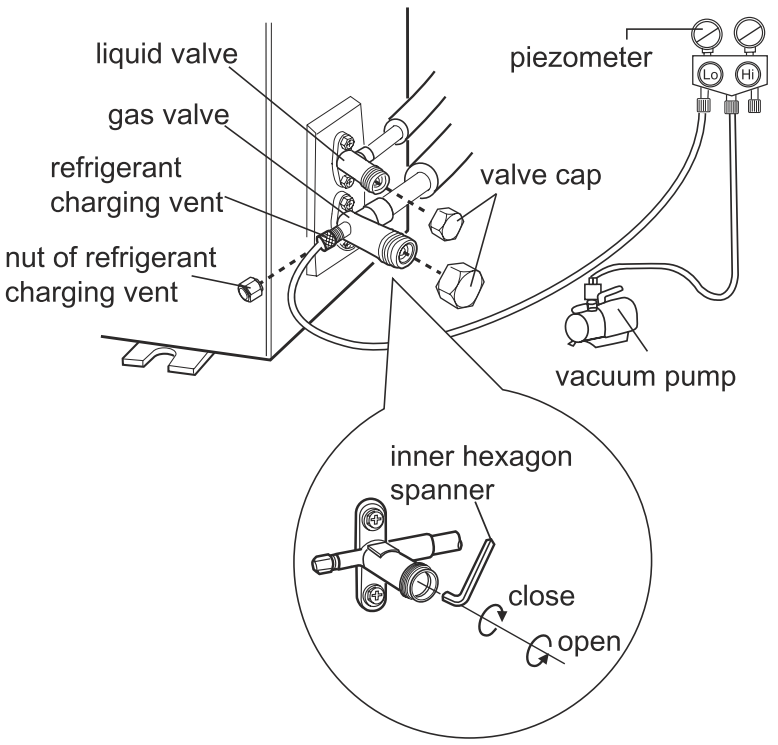
- Slant the drain hose slightly dow-nwards. The drain hose can't be curved, raised and fluctuant, etc.



Vacuum pumping

Use vacuum pump

1. Remove the valve caps on the liquid valve and gas valve and the nut of refrigerant charging vent.
2. Connect the charging hose of piezometer to the refrigerant charging vent of gas valve and then connect the other charging hose to the vacuum pump.
3. Open the piezometer completely and operate for 10-15min to check if the pressure of piezometer remains in -0.1MPa.
4. Close the vacuum pump and maintain this status for 1-2min to check if the pressure of piezometer remains in -0.1MPa. If the pressure decreases, there may be leakage.
5. Remove the piezometer, open the valve core of liquid valve and gas valve completely with inner hexagon spanner.
6. Tighten the screw caps of valves and refrigerant charging vent.
7. Reinstall the handle.



Leakage detection

1. With leakage detector:
   Check if there is leakage with leakage detector.
2. With soap water:
   If leakage detector is not available, please use soap water for leakage detection. Apply soap water at the suspected position and keep the soap water for more than 3min. If there are air bubbles coming out of this position, there's a leakage.

Check after installation

- Check according to the following requirement after finishing installation.

Items to be checked	Possible malfunction
Has the unit been installed firmly?	The unit may drop, shake or emit noise.
Have you done the refrigerant leakage test?	It may cause insufficient cooling/ heating capacity.
Is heat insulation of pipeline sufficient?	It may cause condensation and water dripping.
Is water drained well?	It may cause condensation and water dripping.
Is the voltage of power supply according to the voltage marked on the nameplate?	It may cause malfunction or damaging the parts.
Is electric wiring and pipeline installed correctly?	It may cause malfunction or damaging the parts.
Is the unit grounded securely?	It may cause electric leakage.
Does the power cord follow the specification?	It may cause malfunction or damaging the parts.
Is there any obstruction in the air inlet and outlet?	It may cause insufficient cooling/ heating capacity.
The dust and sundries caused during installation are removed?	It may cause malfunction or damaging the parts.
The gas valve and liquid valve of connection pipe are open completely?	It may cause insufficient cooling/ heating capacit
Is the inlet and outlet of piping hole whether the operation is normal or not.	It may cause insufficient cooling/ heating capacity or waster eletricity.

Test operation

Preparation of test operation

The client approves the air conditioner.

Specify the important notes for air conditioner to the client.

Configuration of connection pipe

1. Standard length of connection pipe
   - 3.5m

2.Min. length of connection pipe is 3m.

3.Max. length of connection pipe and max. height difference.

Cooling capacity	Max length of connec-tion pipe(m)	Max height difference (m)
9000Btu/h (0.75T)	12	7
12000Btu/h (1.0T)	12	7
18000Btu/h (1.5T)	12	7
24000Btu/h (2.0T)	12	7

4. The additional refrigerant oil and refrigerant charging required after prolonging connection pipe
   - After the length of connection pipe is prolonged for 10m at the basis of standard length, you should add 5ml of refrigerant oil for each additional 5m of connection pipe.
   - The calculation method of additional refrigerant charging amount (on the basis of liquid pipe):

     Additional refrigerant charging amount = prolonged length of liquid pipe ? additional refrigerant charging amount per meter
   - Basing on the length of standard pipe, add refrigerant according to the requirement as shown in the table. The additional refrigerant charging amount per meter is different according to the diameter of liquid pipe. See the following sheet.

Configuration of connection pipe

Additional refrigerant charging amount for R22, R407C, R410A, R32

Diameter of connection pipe		Outdoor unit throttle
Liquid pipe(mm)	Gas pipe(mm)	Cooling only(g/m)
Φ6.35	Φ9.52	20
Φ6.35	Φ12.7	20
Φ6.35	Φ15.88	20

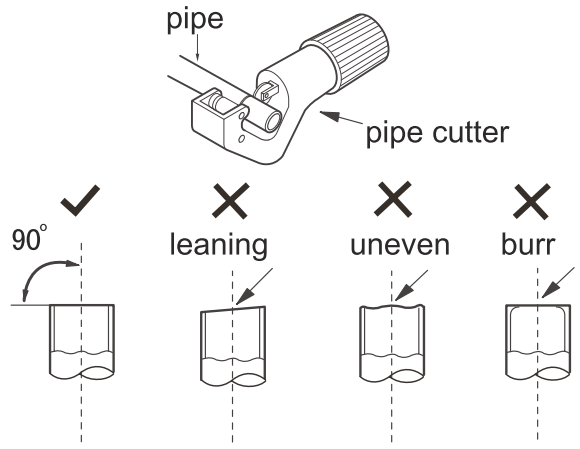
Pipe expanding method

Note:

Improper pipe expanding is the main cause of refrigerant leakage. Please expand the pipe according to the following steps:

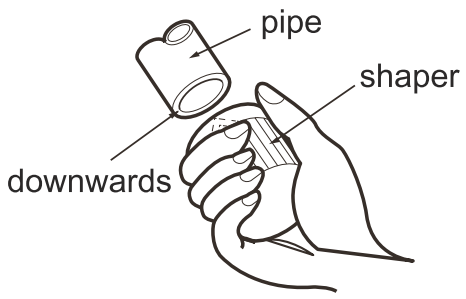
A: Cut the pipe

- Confirm the pipe length according to the distance of indoor unit and outdoor unit.
- Cut the required pipe with pipe cutter.



B: Remove the burrs

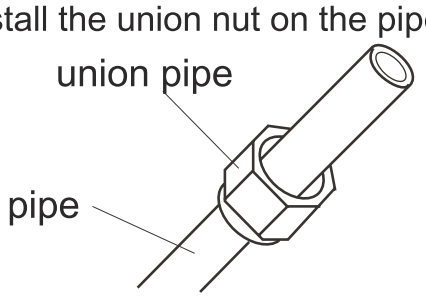
- Remove the burrs with shaper and prevent the burrs from getting into the pipe.



C: Put on suitable insulating pipe

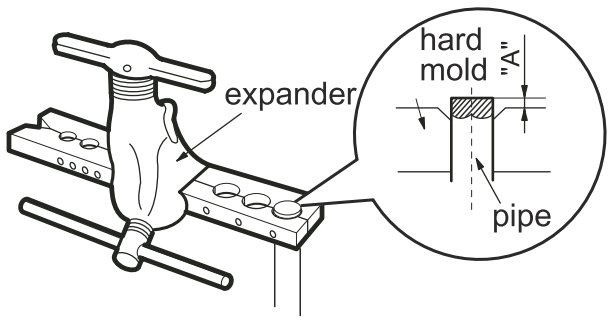
D: Put on the union nut

- Remove the union nut on the indoor connection pipe and outdoor valve; install the union nut on the pipe.



E: Expand the port

- Expand the port with expander.



Note:

- "A" is different according to the diameter, please refer to the sheet below:

Outer diameter (mm)	A(mm)	
	Max	Min
Φ6.35(1/4")	1.3	0.7
Φ9.52(3/8")	1.6	1.0
Φ12.7(1/2")	1.8	1.0
Φ15.88(5/8")	2.4	2.2

F: Inspection

- Check the quality of expanding port. If there is any blemish, expand the port again according to the steps above.

